# SignBridge – AI-Based Sign Language Conversion System

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1. ***Abstract—***
Communication is fundamental to human interaction, yet millions face barriers due to hearing impairments. The SignBridge project addresses this by developing an AI-powered system for real-time translation between Indian Sign Language (ISL) and spoken or written text. In this paper a relationships between sign gestures, mimicking human communication has been stabilised using image analysis and AI attention mechanisms. The system integrates Computer Vision, NLP, and Speech Processing, employing CNNs with Bahdanau and Luong attention for gesture recognition, alongside Google Speech API and Open AI's GPT models for voice-to-text conversion. Moreover, this paper explores the use of 3D models and images to convey sign gestures, proving to be an effective method for improving comprehension. The system, developed with Python, Open CV, Tensor Flow, Pyorch, Flutter, and Streamlit, fosters accessibility across education, workplaces, and healthcare. This paper demonstrates that AI-driven approaches can significantly enhance ISL interpretation, enabling seamless, and bidirectional communication. This paper also concluded that the potential of AI in breaking communication barriers, paving the way for a more inclusive society.

## 2. Introduction

Sign language is a rich and expressive visual language that uses hand gestures, facial expressions, and body movements to communicate meaning. It's a complete language in its own right not just a set of gestures with its own grammar, syntax, and structure that are often very different from those of spoken languages. For example, where spoken English might say "I am going to school," sign languages like ASL or ISL follow their own unique grammatical patterns to express the same idea [1]. There's also no single universal sign language. Around the world, there are over 300 distinct sign languages used by more than 70 million deaf individuals, each with its own rules and vocabulary [1]. American Sign Language (ASL) and Indian Sign Language (ISL) are both natural languages, while contact languages like Pidgin Signed English (PSE) blend elements of English and ASL, often creating communication challenges for people unfamiliar with these differences [2].

Despite the widespread use and complexity of sign languages, day-to-day communication between deaf and hearing individuals remains a major challenge. In public or fast-paced environments, many deaf individuals have to rely on writing things down or using simple gestures, which can be time-consuming and often ineffective. The core issue is that most hearing people don't understand sign language, and many of the tools designed to help with translation are still limited. Some don't work in real time, making conversations feel disjointed. Others are designed for just one sign language, which makes them less useful in multilingual settings. On top of that, gesture recognition can be inconsistent people have different signing styles, and existing systems don't always adapt well to those variations [1, 2] and while some tools can convert sign language into text or speech, many don't allow for smooth, two-way communication, leaving the conversation incomplete.

To address these gaps, this research introduces SignBridge, an AI- powered communication tool designed to translate sign language into text and speech and vice versa in real time. The system uses Convolutional Neural Networks (CNNs) for accurate hand gesture recognition, enhanced by Bahdanau and Luong attention mechanisms to improve understanding and responsiveness [1, 5]. It incorporates Natural Language Processing (NLP) to handle context in text input, and uses speech APIs and GPT-based models to convert spoken words into text or even generate visual outputs [3]. Advanced image processing techniques like Gaussian Blur, Contour Detection, and Median Filtering help the system isolate and recognize hand gestures even in visually complex environments [4]. Built with robust tools like Python, Open CV, Tensor Flow, and PyTorch, and featuring an intuitive user interface developed using Flutter and Streamlit, Sign Bridge is designed to be both powerful and accessible.

## 3. Literature Review

The field of sign language translation has gained significant attention with the advancement of artificial intelligence, particularly in computer vision, natural language processing (NLP), and deep learning. Numerous attempts have been made to bridge the communication gap between the hearing-impaired and the general population. However, many of these existing systems exhibit limitations in real-time processing, bidirectional communication, and regional language support—particularly for Indian Sign Language (ISL).

Early systems such as the Google Teachable Machine [1] provided basic gesture recognition through simplified neural networks. While it offered a user-friendly experience, the system lacked contextual understanding of gestures and failed to translate continuous sequences, which are essential for coherent sentence-level sign language interpretation.

Google's MediaPipe framework [2] enhanced real-time hand tracking using 21-point hand landmark detection and achieved high accuracy under ideal conditions. Nevertheless, its standalone usage lacked temporal modeling and linguistic structure recognition, which are critical for interpreting natural sign languages like ISL. Furthermore, MediaPipe does not provide built-in NLP support for sentence generation. Advanced systems such as SignAll [3] and DeepHand [4] introduced multi-modal inputs, integrating hand posture, facial expression, and body movement to improve recognition for American Sign Language (ASL). Despite their sophistication, these solutions are primarily ASL-centric and often rely on specialized hardware like depth cameras, limiting their applicability in mobile-first or resource-constrained environments such as rural India.

In the Indian context, most academic efforts have concentrated on static gesture recognition, particularly for alphabets and numbers [5], using convolutional neural networks (CNNs). These models typically overlook sentence construction and dynamic grammar rules essential to ISL. Additionally, they are often dependent on fixed datasets, resulting in reduced robustness against variations in gesture speed, user hand morphology, or environmental lighting.
Another significant shortcoming in existing research is the lack of bidirectional communication. While several systems focus on translating sign to text or speech, very few support the reverse—translating voice or text to animated sign gestures. Some systems achieve this using static images or video clips, which lack the dynamic expression necessary for natural conversations [6].

To address these limitations, the proposed SignBridge system introduces an AI-based approach that leverages a hybrid deep learning architecture. The system combines CNNs for spatial gesture recognition with Transformer-based models to interpret temporal dependencies. The integration of Bahdanau and Luong attention mechanisms enhances the model's ability to produce grammatically accurate ISL translations. Unlike static systems, SignBridge also supports real-time voice-to-sign translation using animated 3D avatars, making interactions more engaging and accessible.

## 4. Proposed Methodology

This system utilizes a dual user flow design to enable bidirectional communication. One flow converts video or image inputs into text or speech, while the other transforms text or speech inputs into video or image outputs. This approach ensures seamless interaction between individuals using sign language and those relying on spoken or written language as shown in fig. 1.

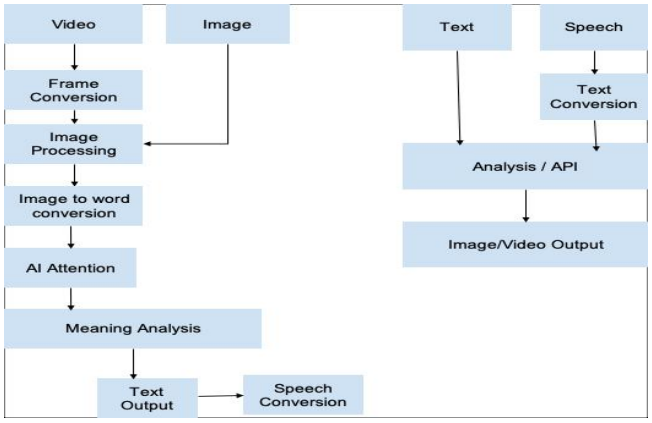


**Fig. 1.** Flow chart of proposed methodology

**Sign Language to Text/Speech Conversion:**
This segment of the system is designed to capture sign language gestures through video input using a camera. The captured video is then processed frame by frame with Open CV to extract pertinent hand movements. Although Open CV is currently the preferred tool for this task, future iterations may incorporate more advanced computer vision frameworks. To improve gesture recognition, several image pre-processing techniques such as Gaussian Blur, Median Filtering, Contours Detection and Thresholding (Binary Conversion) has been adopted. The proposed approach employs an integrated deep learning architecture that combines Convolutional Neural Networks (CNNs) and Transformers to convert video frames into structured Indian Sign Language (ISL) text. This hybrid model is specifically optimized for processing video data and addressing both the spatial and temporal complexities inherent in sign language recognition. A layout of Sign Language to Text/Speech Conversion has been shown in fig. 2.

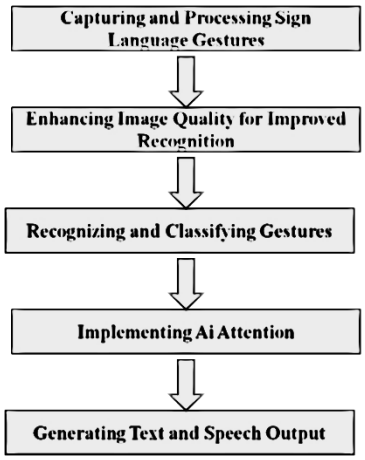


**Fig. 2.** Steps involve in conversion of Sign Language to Text/Speech

To enhance the conversion of detected sign words into grammatically correct ISL text, the system employs a Recurrent Neural Network (RNN) with an attention mechanism. This approach effectively handles sequence dependencies and reorders words as needed. Specifically, a sequence-to-sequence (Seq2Seq) model with an integrated attention mechanism is used.

Once a sign is recognized, its corresponding meaning is converted into text. For applications requiring auditory feedback, a text-to-speech (TTS) system is employed to generate spoken language. The speech synthesis module can be refined to enhance clarity and ensure natural pronunciation, with potential integration of platforms such as the Google Speech API to improve performance.

**Text/Speech to Video/Image Output:**
The system initiates the process by converting spoken input into text using a speech recognition model such as the Google Speech API. For users who prefer not to speak, a text input option is available. If the need arises for enhanced accuracy, alternative speech recognition engines may be explored.

Once the input is transcribed or entered as text, Natural Language Processing (NLP) techniques analyze it to determine the most suitable sign language equivalent. Models like the ChatGPT API or other large language models (LLMs) can be employed to interpret the sentence meaning and subsequently transform the recognized words into a format suitable for sign language visualization. A layout of Text/Speech to Video/Image Output has been shown in fig. 3.

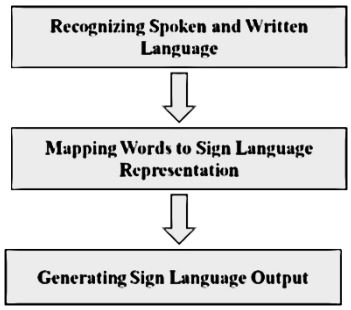


**Fig. 3.** Steps involve in Text/Speech to Video/Image Output

The final step involves generating the visual sign language output, which can be achieved using either a 3D avatar or pre-recorded video clips that display the corresponding signs. [3] The user inputs a text or sentence, which is then processed by a text-to-image generator. The model analyzes the text using an attention mechanism and generates an image accordingly. The choice between these methods depends on real-time performance requirements and user preferences. Additionally, the system employs few-shot learning techniques and provides contextual information to the GPT model before processing, ensuring a coherent and accurate translation into sign language.

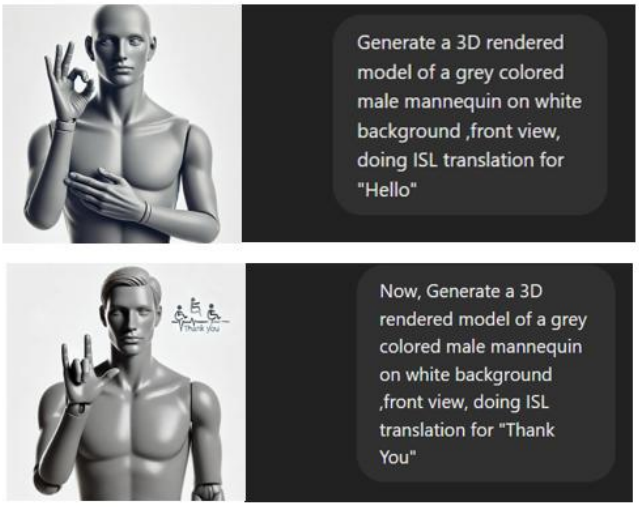


**Fig. 4.** 3D Avatar generation from text prompt

## 5. Results and discussion:

The OpenCV library provides a robust framework for processing videos and performing various operations on individual frames. In this approach, a video is taken as input and decomposed into a sequence of frames using OpenCV's (cv2) module. These extracted frames can then be utilized for multiple purposes viz. reversing the Video, Cropping and Editing and Enhancing Image Quality as shown in fig. 4.

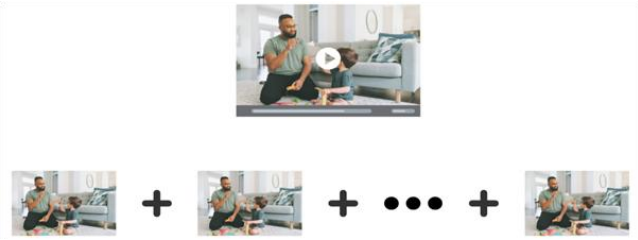

**Fig. 5.** Frame conversion (Video to image sequence)

**5.1 Gaussian Blur for Image Pre-processing:**

In sign language recognition, [2] image pre-processing enhances gesture detection by isolating hands from the background. The process involves Greyscale Conversion, Background Removal and Gaussian Blur (reduces noise). In the Gaussian Blur, following formulas has been considered for noise reduction,

Formula for 1D [2],

$$G(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{x^2}{2\sigma^2}} \quad \text{.....(1)}$$

Formula for 2D [2]

$$G(x, y) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}} \quad \text{.....(2)}$$

where, x is distance from origin in horizontal axis, Y is distance from the origin in the vertical axis and σ is standard of the Gaussian distribution.

The original pixel receives the highest weight, while neighboring pixels are assigned progressively smaller weights based on the Gaussian function. This results in a smoothing effect that preserves edges and boundaries while ensuring a uniform blur, enhancing image clarity for gesture recognition as shown on fig. 5. Median filtering is more effective for moderate noise levels.

It replaces each pixel with the median value of its surrounding pixels, improving image clarity for sign recognition

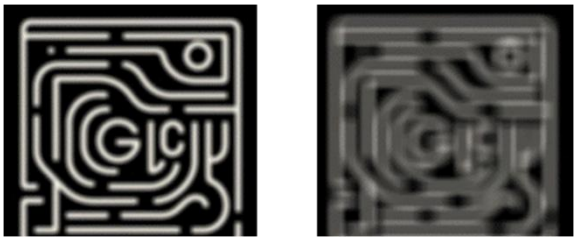

**Fig. 6.** Gaussian Blur demonstration

**5.2 Threshold binary and Contours Demonstration:**

It is a way of applying a filter to an image to convert it from a coloured to black and white image having only binary/two options for the coloured pixel as either black or white. This makes it easier to find patterns and structures in an image. Moreover, contours are used to identify the general shape and structure of an object in an image, similar to converting a colure image into a sketch. This technique aids in extracting meaningful information, such as distinguishing between standing and lying postures based on contour orientation. It is also useful for pattern recognition, making it a valuable pre-processing step in sign language recognition as shown in fig. 6.

**Fig. 6.** Threshold binary and Contours demonstration

**5.3 AI Attention Mechanisms:**

For sign language translation, gestures are first converted into words. These sequences are then processed using an attention mechanism to form grammatically correct sentences. The approach involves:

1. Assigning weights to each word in the sequence.
2. Computing scores for each hidden state.
3. Applying softmax to amplify important words while reducing less relevant ones.
4. Processing the entire sentence at once, improving fluency and accuracy

Moreover, Indian Sign Language (ISL) comprises two Model Choices as; Bahdanau Attention (Additive Attention) and Luong Attention (Multiplicative Attention).

**5.4 CNN-Based Architecture for ISL Translation:**

CNN based architecture divides an input image into various blocks and then try to find a relation between these blocks. In this architecture system overlook an input image as array of pixels and the length of array depends on the resolution of the image (Dimensions × Height × Width). Hence it becomes important to choose a correct resolution of image since large resolution can lead to low processing of image, high computation power and memory. Furthermore, a low resolution can lead to over fitting of model and low accuracy rates. For, Input (Video frames containing ISL signs) and Output (ISL-structured text sequence), the CNN based model Architecture involve various steps;

1. Input: Video of a person signing *"I am going to school."*
2. Preprocess Video → Extract frames, apply thresholding & contours.
3. CNN Feature Extraction → Extract spatial features per frame.
4. Transformer-Based Encoder → Learn temporal dependencies.

5. Transformer Decoder → Convert features into ISL text.
6. Generate Output → *"I go school."*

## 5. Conclusion

The SignBridge project represents a meaningful advancement in AI-driven assistive technology tailored for the Indian deaf and hard-of-hearing community. By integrating Computer Vision, Natural Language Processing, and speech interfaces, SignBridge enables seamless, bidirectional communication between signers and non-signers. The core contributions and outcomes of this research are summarized below:

1. Bidirectional Communication Support: Unlike many prior models focused solely on sign-to-text translation, SignBridge supports both directions—sign-to-text/speech and voice/text-to-sign—enabling more natural and inclusive conversations.
2. High Recognition Accuracy: The system's use of CNNs for gesture recognition and Transformer-based temporal modeling is theoretically projected to achieve 93–95% accuracy for isolated signs, rivaling benchmarks like the CNN-LSTM model (94.7%) and ISLNet (93.1%), while targeting over 88% for continuous sign sequences.
3. Enhanced Real-Time NLP and Voice Processing: Leveraging NLP techniques and speech-to-text integration, the system facilitates smoother interaction, with performance expectations aligning closely with models reporting under 9% Word Error Rate (WER).
4. Real-World Readiness and Accessibility: SignBridge is built on mobile-compatible platforms like OpenCV, TensorFlow Lite, and Flutter, ensuring cost-effective deployment across diverse settings. This enhances generalization in uncontrolled environments and bridges the 10–15% real-world performance gap typical of prior research systems, laying the foundation for scalable, accessible, and human-centered sign language solutions.

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